C 43201 **(Pages : 2)** **Name.....**

Reg. No.....

SECOND SEMESTER (CBCSS—UG) DEGREE EXAMINATION
APRIL 2023

Physics/Applied Physics

PHY 2C 02—OPTICS, LASER, ELECTRONICS

(2019—2022 Admissions)

Time : Two Hours Maximum : 60 Marks

Section A (Short Answer Type)

Answer **all** *questions in two* or *three sentences.*
Each correct answer carries a maximum of 2 *marks*

1. What are the conditions for obtaining sustained interference ?
2. What is the origin for colors exhibited by thin films ?
3. What is diffraction? What is the condition for obtaining a noticeable diffraction effect ?
4. Mention two features of Fresnel diffraction.
5. In the Fraunhoffer diffraction at a single slit, draw the intensity distribution.
6. Distinguish between transmission and reflection gratings. Give an expression for the principal maximum of order, say *n,* in the diffraction spectrum.
7. Write any *two* differences between interference and diffraction patterns.
8. What do you mean by double refraction ? Distinguish between positive and negative crystals.
9. What do you mean by a plane polarized light ? When a plane polarized light is passed through a polarizer, how many times will it be extinguished in one full rotation of the polarizer ?
10. What do you mean by a filter circuit ? Explain the construction of a π-section filter.
11. What do the Einstein's co-efficients stand for ?
12. List any three essential characteristics of spontaneous emission.

(Ceiling 20)

Turn over

Section B (Paragraph/Problem Type)

Answer **all** *questions in* **a paragraph** *of about* **half a page to one page**.
Each correct answer carries a maximum of 5 marks.

13. In Newton's rings experiment with reflected light, the diameter of 15th ring is 0.6 cm and that of 5th ring is 0.3 cm. If the radius of the plano-convex lens is 100 cm, what is the wavelength of light used in the experiment. ?

14. In Fraunhoffer diffraction pattern due to a narrow slit, a screen is placed 2 m away from the lens to obtain the pattern. If the wavelength of light used is 5×10^{-5} cm and the first minimum lie at 5 mm on either side of the central maximum, what is the slit width ?

15. Using suitable figures, explain the term optical activity.

16. Calculate the minimum thickness of a calcite plate which would convert a plane polarized light to circularly polarized. Given, the wavelength of light λ =5890 Å, refractive index of ordinary ray = 1.658 and the refractive index of extra ordinary ray =1.486.

17. A transistor has α = 0.98, I_B = 100 μA and I_{CBO} = 5 μA. Determine emitter current, collector current and the amplification factor β.

18. Give the truth table of an exclusive OR gate. How will you construct it using basic gates ?

19. Give the construction and the basic details of a He-Ne laser.

(Ceiling 30)

Section C (Essay Type)

Essays.
Answer in about **two pages**.
Answer any one question.
Answer carries 10 *marks.*

20. Give the analytical treatment of interference of two sinusoidal waves. Discuss the conditions for maximum and minimum intensity. Plot the energy distribution as a function of phase angle.

21. Discuss the working principle of a centre-tapped full wave bridge rectifier using suitable figures. Obtain an expression for its efficiency.

(1 × 10 = 10 marks)